

eNSP ISP Lab — Detailed Documentation & Ready-to-Paste Configs

Purpose: Build a realistic, lab-scale ISP environment in Huawei eNSP showing OSPF core, MPLS LDP, iBGP between PEs, eBGP with customers and a simulated Internet stub, plus a basic MPLS-VPN (VRF) for a customer with two sites.

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1) Topology (ASCII)

```
[CE1 LAN]--(CE1 AR1220)--[172.16.1.2] === [172.16.1.1]--(PE1 AR2220)--
[10.0.12.1]

                                     |
                               [10.0.12.2]
                                     |
                               (P1 AR2220)
                                     |
    [10.0.23.1]--(to PE2 AR2220)--[10.0.23.2]
                                     |
                               [172.16.2.1]--(PE2)--[172.16.2.2]--(CE2
AR1220)--[CE2 LAN]

                                     [10.0.24.1]--(P1)--[10.0.24.2]--(PE3 AR2220)--
    [200.1.1.1]--(Internet AR1220)--[200.1.1.2]

Loopbacks: PE1=1.1.1.1, P1=1.1.1.2, PE2=1.1.1.3, PE3=1.1.1.4
Internet loopback: 8.8.8.8/32
```

2) Devices & recommended models

- CE1: **AR1220** (customer edge)
- CE2: **AR1220**
- PE1: **AR2220** (provider edge)
- P1: **AR2220** (core P router)
- PE2: **AR2220**
- PE3: **AR2220** (peering with Internet)
- Internet (stub): **AR1220**
- Optional access switches: **S5700**

Use AR2220/AR3260 for routers that must support MPLS, VRF, and BGP. AR1220 works well for CE and simple stub routers.

3) Addressing plan & ASNs

ASNs - ISP core (PE/P): 65000 - Customers (CE1/CE2): 65010 - Internet stub: 65001

Loopbacks (PE/P IDs) - PE1: 1.1.1.1/32 - P1: 1.1.1.2/32 - PE2: 1.1.1.3/32 - PE3: 1.1.1.4/32 - Internet Loopback: 8.8.8.8/32

Point-to-point (/30) subnets - CE1 <-> PE1: 172.16.1.0/30 (CE1 .2, PE1 .1) - PE1 <-> P1: 10.0.12.0/30 (PE1 .1, P1 .2) - P1 <-> PE2: 10.0.23.0/30 (P1 .1, PE2 .2) - PE2 <-> CE2: 172.16.2.0/30 (PE2 .1, CE2 .2) - P1 <-> PE3: 10.0.24.0/30 (P1 .1, PE3 .2) - PE3 <-> Internet: 200.1.1.0/30 (PE3 .1, Internet .2)

Customer LANs - CE1 LAN: 192.168.10.0/24 (CE1 gateway 192.168.10.1) - CE2 LAN: 192.168.20.0/24 (CE2 gateway 192.168.20.1)

4) Design overview

- **IGP inside ISP**: OSPF area 0 across PE1-P1-PE2-PE3 (loopbacks + P2P links).
- **MPLS**: MPLS LDP enabled on PE-P-PE links to build LSPs for label switched forwarding.
- **iBGP**: PEs peer using loopbacks (1.1.1.x) inside AS65000 — either full mesh or via route-reflector.
- **eBGP**: CE1/CE2 run eBGP to their respective PEs (AS65010 ↔ 65000). Internet stub runs eBGP with PE3 (65001 ↔ 65000).
- **VPN / VRF**: `ip vpn-instance CUST-A` on PE1/PE2 (RD/RT = 65000:10) — customer site prefixes exchanged as VPNv4 and kept separated from global table.
- **Internet access**: The Internet stub advertises 8.8.8.8/32. Two lab-friendly options to give customers Internet:
- PE3 injects a default into the CUST-A vpn-instance (advertised to other PEs).

- Or use RT-based import of Internet/global default into the VRF (more production-like; needs route-policy).

5) Step-by-step build instructions

1. Create devices in eNSP and connect links according to the IP plan.
2. Configure interfaces and loopbacks on all routers.
3. Configure OSPF on PE1/P1/PE2/PE3 so loopbacks and P2P links are in area 0.
4. Enable `mpls` and `mpls ldp` on PE-P-PE links.
5. Configure iBGP on PEs using loopbacks. Ensure `peer x.x.x.x connect-interface LoopBack0` if your image requires.
6. Configure eBGP on CE1/CE2 and on PE3 ↔ Internet.
7. Configure `ip vpn-instance CUST-A` on PE1 and PE2 with same RD/RT and BGP vpn-instance sections to peer with CEs.
8. Provide a default route in the `CUST-A` vpn-instance (e.g., on PE3 advertise a static default into vpn-instance) so customers can reach Internet.
9. Verify: OSPF neighbors, MPLS LDP adjacencies, iBGP sessions, eBGP with CEs/Internet, VPN route exchange, and ping/traceroute tests.

6) Full ready-to-paste configurations

These are the recommended configs for each device. If your eNSP VRP image uses slightly different syntax for vpn-instance BGP sections or interface names, adjust those lines accordingly.

CE1 (AR1220) — AS 65010

```
sysname CE1

interface GigabitEthernet0/0/0
description LAN-to-users
ip address 192.168.10.1 255.255.255.0

interface GigabitEthernet0/0/1
description to-PE1
ip address 172.16.1.2 255.255.255.252

# eBGP to PE1
bgp 65010
peer 172.16.1.1 as-number 65000
network 192.168.10.0 255.255.255.0
```

```
stelnet server enable
return
```

CE2 (AR1220) — AS 65010

```
sysname CE2

interface GigabitEthernet0/0/0
  description LAN-to-users
  ip address 192.168.20.1 255.255.255.0

interface GigabitEthernet0/0/1
  description to-PE2
  ip address 172.16.2.2 255.255.255.252

# eBGP to PE2
bgp 65010
  peer 172.16.2.1 as-number 65000
  network 192.168.20.0 255.255.255.0

stelnet server enable
return
```

PE1 (AR2220) — ISP AS 65000 (PE facing CE1 & core)

```
sysname PE1

# Interfaces
interface LoopBack0
  ip address 1.1.1.1 255.255.255.255

interface GigabitEthernet0/0/0
  description to-CE1
  ip address 172.16.1.1 255.255.255.252
  # bind the customer-side interface to vpn-instance (VRF)
  ip binding vpn-instance CUST-A

interface GigabitEthernet0/0/1
  description to-P1
  ip address 10.0.12.1 255.255.255.252
  mpls
```

```

# OSPF (IGP) - area 0 covers P2P links & loopbacks
ospf 1
 area 0.0.0.0
  network 10.0.12.0 0.0.0.3
  network 1.1.1.1 0.0.0.0

# MPLS LDP
mpls
 mpls lsr-id 1.1.1.1
 mpls ldp
  enable

# iBGP with other PEs (loopbacks)
bgp 65000
 peer 1.1.1.3 as-number 65000
 peer 1.1.1.4 as-number 65000

# VPN-instance (VRF) for Customer A
ip vpn-instance CUST-A
 description Customer-A-VRF
 route-distinguisher 65000:10
 vpn-target 65000:10 both

# BGP vpn-instance to talk to CE1 inside VRF
bgp 65000
 ipv4-family vpn-instance CUST-A
  peer 172.16.1.2 as-number 65010
  undo synchronization

telnet server enable
return

```

P1 (AR2220) — Core P router (OSPF + MPLS)

```

sysname P1

interface LoopBack0
 ip address 1.1.1.2 255.255.255.255

interface GigabitEthernet0/0/0
 description to-PE1
 ip address 10.0.12.2 255.255.255.252
 mpls

interface GigabitEthernet0/0/1

```

```
description to-PE2
ip address 10.0.23.1 255.255.255.252
mpls

interface GigabitEthernet0/0/2
description to-PE3
ip address 10.0.24.1 255.255.255.252
mpls

# OSPF
ospf 1
area 0.0.0.0
network 10.0.12.0 0.0.0.3
network 10.0.23.0 0.0.0.3
network 10.0.24.0 0.0.0.3
network 1.1.1.2 0.0.0.0

# MPLS LDP
mpls
mpls lsr-id 1.1.1.2
mpls ldp
enable

stelnet server enable
return
```

PE2 (AR2220) — ISP AS 65000 (PE facing CE2 & core)

```
sysname PE2

interface LoopBack0
ip address 1.1.1.3 255.255.255.255

interface GigabitEthernet0/0/0
description to-P1
ip address 10.0.23.2 255.255.255.252
mpls

interface GigabitEthernet0/0/1
description to-CE2
ip address 172.16.2.1 255.255.255.252
ip binding vpn-instance CUST-A

# OSPF
ospf 1
```

```
area 0.0.0.0
 network 10.0.23.0 0.0.0.3
 network 1.1.1.3 0.0.0.0

# MPLS LDP
mpls
mpls lsr-id 1.1.1.3
mpls ldp
 enable

# iBGP with other PEs
bgp 65000
 peer 1.1.1.1 as-number 65000
 peer 1.1.1.4 as-number 65000

# VPN-instance (VRF) for Customer A
ip vpn-instance CUST-A
 description Customer-A-VRF
 route-distinguisher 65000:10
 vpn-target 65000:10 both

# BGP vpn-instance to talk to CE2 inside VRF
bgp 65000
 ipv4-family vpn-instance CUST-A
  peer 172.16.2.2 as-number 65010
  undo synchronization

stelnet server enable
return
```

PE3 (AR2220) — ISP edge to Internet (AS 65000)

```
sysname PE3

interface LoopBack0
 ip address 1.1.1.4 255.255.255.255

interface GigabitEthernet0/0/0
 description to-Internet
 ip address 200.1.1.1 255.255.255.252

interface GigabitEthernet0/0/1
 description to-P1
 ip address 10.0.24.2 255.255.255.252
mpls
```

```
# OSPF
ospf 1
 area 0.0.0.0
  network 10.0.24.0 0.0.0.3
  network 1.1.1.4 0.0.0.0

# MPLS LDP
mpls
 mpls lsr-id 1.1.1.4
 mpls ldp
 enable

# iBGP to other PEs
bgp 65000
 peer 1.1.1.1 as-number 65000
 peer 1.1.1.3 as-number 65000

# eBGP to Internet stub
bgp 65000
 peer 200.1.1.2 as-number 65001

# advertise default into CUST-A vpn-instance by injecting static route or using
import-route
network 0.0.0.0 0.0.0.0

stelnet server enable
return
```

Internet Router (AR1220) — AS 65001 (stub)

```
sysname Internet

interface LoopBack0
 ip address 8.8.8.8 255.255.255.255

interface GigabitEthernet0/0/0
 description to-PE3
 ip address 200.1.1.2 255.255.255.252

# Advertise test prefix (8.8.8.8)
bgp 65001
 peer 200.1.1.1 as-number 65000
 network 8.8.8.8 255.255.255.255
```



```
stelnet server enable  
return
```

7) Verification commands & expected outputs (examples)

Run these commands and check for the described status.

OSPF checks (on PE1/P1/PE2/PE3)

```
display ospf peer
```

- Expect neighbors in **Full** state for all P2P links. - Loopbacks of each router visible in OSPF LSDB.

MPLS LDP (PE & P)

```
display mpls ldp peer
```

- Expect adjacent LDP peers with session **Operational**.

iBGP checks (on PEs)

```
display bgp peer  
display bgp vpnv4 all routing-table
```

- iBGP neighbors **1.1.1.x** should be **Established**. - VPNv4 table should include customer prefixes (192.168.10.0/24, 192.168.20.0/24).

CE eBGP checks

On CE1:

```
display bgp peer  
display ip routing-table  
ping 8.8.8.8
```

- eBGP neighbor to PE1 **Established**. - Default route or learned route should exist pointing to PE1. - Ping 8.8.8.8 should succeed once PE3 advertises default into the VPN.

Traceroute across VRF

```
tracert vpn-instance CUST-A 192.168.20.1
```

- Should show labeled hops across P1 from PE1 to PE2.

8) Common troubleshooting

- **eBGP not established:** check AS numbers, TTL (eBGP uses TTL 1 for directly connected), interface IPs, and ACLs.
- **iBGP not established:** ensure loopbacks are reachable via OSPF and that `peer x.x.x.x connect-interface LoopBack0` is used if required.
- **No VPN route at other PE:** verify RD/RT values match, `bgp ipv4-family vpn-instance CUST-A` sections exist, and `import-route` if using static default injection.
- **CE cannot ping Internet:** ensure default exists in VRF (`display ip vpn-instance CUST-A routing-table`) and that PE3 is exporting default via BGP vpnv4.

9) Optional enhancements

- **Route Reflector:** deploy an RR to avoid full-mesh on many PEs.
- **NAT:** add NAT on PE3 if customer uses private IPs and Internet requires PAT.
- **QoS:** use traffic shaping/policers per VRF on PE interfaces.
- **Monitoring:** SNMP, NetFlow on PEs for traffic visibility.

Appendix: Quick fixes (give CE1 Internet)

If CE1 cannot reach Internet, execute this on **PE3** to inject a default into the VPN (lab-friendly):

```
ip vpn-instance CUST-A
 route-distinguisher 65000:10
 vpn-target 65000:10 both
quit

ip route-static vpn-instance CUST-A 0.0.0.0 0.0.0.0 200.1.1.2

bgp 65000
 ipv4-family vpn-instance CUST-A
  import-route static
  undo synchronization
quit
```

End of document

If you want: I can also - generate a **PNG diagram** of the topology (for copy into eNSP), or - produce a single **paste-sequence script** (a file where each router's config is in a block you can paste sequentially), or - customise the lab (more customers, VRFs, NAT, RR configuration).

Tell me which extra item you want next.